Science practice booklet: Grade 9 expectations

Grade 9 · Practice booklet

Name: _____

What this booklet is

This is general practice written for Science at Grade 9 level, following the Next Generation Science Standards standards for that grade. It is not made from your child's MAP questions, and it does not replace the classroom. Use it as a guide to what a child is expected to know and do at this level.

How to use it

- Do one section at a time, about 10 minutes, 3 days a week. Stop while it is still enjoyable.
- Read each question all the way through, twice, before answering. Underline the key words and say what is being asked.
- Take your time. A careful, correct answer is worth more than a fast one.
- Use scrap paper and show the working. Then check the answer before moving on.
- Ask your child to explain how they got an answer, even when it is right.
- If a section feels too hard, do the section before it again, or ask the teacher. That is part of learning, not a problem.

Section 1: Living things

Standards practised: HS-LS1-1, HS-LS1-3, HS-LS1-4, HS-LS1-5, HS-LS1-7, HS-LS3-1, HS-LS3-3

A worked example

One strand of a DNA molecule has the bases A T G G C A. Write the bases of the strand that pairs with it. Then say how many 3-base codons the original strand holds.

1. DNA bases pair in a fixed way: A pairs with T, and C pairs with G.
2. A T G G C A pairs with T A C C G T.
3. The original strand has 6 bases. Each codon is 3 bases, so 6 ÷ 3 = 2 codons.

The pairing strand is T A C C G T; the original strand holds 2 codons.

Questions

1. Which part of a cell is the main place where energy from food is released?

   A. Nucleus

   B. Mitochondria

   C. Cell wall

   D. Ribosome

2. Give two structures found in a plant cell but not in an animal cell, and say what each one does.

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3. What is the job of the cell membrane? Why does a cell need it?

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4. One DNA strand reads C G T A A G. Write the strand that pairs with it.

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5. A sample of DNA has 30 percent adenine (A). What percentage of the bases is guanine (G)?

   A. 30 percent

   B. 40 percent

   C. 20 percent

   D. 60 percent

6. A gene is a section of DNA. What does a gene code for? If a gene region has 300 bases and every 3 bases code for one amino acid, how many amino acids are in the protein?

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7. A scrape on a student's knee heals as skin cells divide by mitosis. When one skin cell divides, what results?

   A. Two cells, each with the same chromosomes as the original

   B. Two cells, each with half the chromosomes

   C. Four cells, each with half the chromosomes

   D. Four cells, each with the same chromosomes

8. A lab technician in Abu Dhabi watches one cell divide by mitosis, and each new cell divides again, for 4 rounds of division in total. How many cells are there at the end?

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9. Give two reasons why the cells of a young date palm and of a growing child need to divide.

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10. Which equation shows cellular respiration?

    A. Carbon dioxide + water + energy from light makes glucose + oxygen

    B. Glucose + carbon dioxide makes oxygen + water + energy released

    C. Oxygen + water makes glucose + carbon dioxide

    D. Glucose + oxygen makes carbon dioxide + water + energy released

11. In a school garden, tall (T) pea plants are dominant over short (t). Two tall plants, both with the genes Tt, are crossed. What fraction of their offspring are expected to be short?

    A. One quarter

    B. One half

    C. Three quarters

    D. None of them

12. On a hot day in the UAE the body sweats. Explain how this helps keep the body temperature steady, and name this kind of control.

Section 2: Earth and space

Standards practised: HS-ESS2-2, HS-ESS2-4, HS-ESS2-6, HS-ESS3-1, HS-ESS3-3

A worked example

A car gives off about 120 g of carbon dioxide for every kilometre it travels. A family drives 15 000 km in a year. How many kilograms of carbon dioxide does the car give off in that year?

1. Total in grams = 120 g × 15 000 = 1 800 000 g.
2. There are 1000 g in 1 kg, so divide by 1000: 1 800 000 ÷ 1000 = 1800 kg.
3. This carbon dioxide goes into the atmosphere, where it adds to the greenhouse effect.

1800 kg (1.8 tonnes) of carbon dioxide in a year.

Questions

1. Which pair matches an Earth system with a correct example from the UAE?

   A. Hydrosphere: the Hajar mountains

   B. Atmosphere: mangrove trees

   C. Biosphere: the water of the Arabian Gulf

   D. Geosphere: the sand dunes of the desert

2. Mangrove trees grow along the coast of the UAE. Describe how they connect three Earth systems.

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3. Explain how the Sun's energy drives the water cycle.

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4. On a UAE beach in the afternoon, the wind usually blows from the sea towards the land. What is the reason?

   A. Air over the sea rises because water heats faster

   B. Air over the land rises because land heats faster than water

   C. The Moon pulls air towards the land

   D. The Earth's spin pushes air onto the land

5. Which statement describes the climate of the UAE rather than the weather on one day?

   A. A dust storm reduced the view in Al Ain on Tuesday

   B. It rained in Fujairah yesterday evening

   C. Summers are usually hot and humid near the coast

   D. This morning was foggy in Abu Dhabi

6. Describe two ways carbon dioxide gets into the atmosphere and one way it is removed.

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7. Explain the greenhouse effect in your own words. Why is it needed for life on Earth?

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8. Why does burning more fossil fuels make Earth's average temperature rise?

   A. It releases extra carbon dioxide, which traps more heat

   B. It makes the Sun give off more heat

   C. It makes a hole in the ozone layer that lets in the Sun's heat

   D. It removes oxygen, and air with less oxygen holds more heat

9. A home uses 600 kWh of electricity in a 30 day month. Solar panels on its roof make 5 kWh each day. What percentage of the home's electricity do the panels supply?

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10. Give one reason why solar energy is called renewable and coal is called non-renewable.

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11. The Arabian Gulf takes in carbon dioxide from the air. What is one harmful result of this for coral reefs and shellfish?

    A. The water becomes saltier, so corals lose water

    B. The water becomes more acidic, so shells and skeletons are harder to build

    C. The water becomes less acidic, so shells grow too fast

    D. The water becomes clearer, so corals get too much light

12. Ice reflects much more sunlight than dark ocean water. Explain how melting sea ice can lead to even more warming.

Section 3: Matter, forces and energy

Standards practised: HS-PS1-1, HS-PS1-2, HS-PS1-5, HS-PS1-7

A worked example

Hydrogen gas burns in oxygen to make water. The unbalanced equation is H2 + O2 → H2O. Balance it.

1. Count the atoms on each side: left has 2 H and 2 O, right has 2 H and 1 O. The oxygen is not balanced.
2. Put 2 in front of H2O: now the right side has 4 H and 2 O.
3. Put 2 in front of H2: now the left side has 4 H and 2 O. Both sides match.

2 H2 + O2 → 2 H2O

Questions

1. Which statement about the particles in an atom is correct?

   A. Protons are positive and are found in the nucleus

   B. Neutrons are positive and are found in the nucleus

   C. Electrons are negative and are found in the nucleus

   D. Protons are negative and move around the nucleus

2. A sodium atom has atomic number 11 and mass number 23. How many protons, neutrons and electrons does the neutral atom have?

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3. Chlorine has atomic number 17, so its electrons are arranged in shells as 2, 8, 7. How many valence electrons does a chlorine atom have?

   A. 8

   B. 17

   C. 7

   D. 2

4. Lithium, sodium and potassium are in the same group of the periodic table. Why do they react in similar ways?

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5. Helium and neon are very unreactive gases. Use their outer electrons to explain why.

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6. How do sodium and chlorine atoms form sodium chloride (table salt)?

   A. The two atoms share a pair of electrons

   B. Chlorine gives an electron to sodium

   C. Both atoms lose an electron

   D. Sodium gives an electron to chlorine, and the ions attract

7. A desalination plant in the UAE turns seawater into fresh water. In a water molecule (H_2O) the atoms share electrons. What is this kind of bond called, and how is it different from an ionic bond?

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8. Which equation for the burning of magnesium is balanced?

   A. $Mg + O_2 \rightarrow 2\ MgO$

   B. $2\ Mg + 2\ O_2 \rightarrow 2\ MgO$

   C. $2\ Mg + O_2 \rightarrow 2\ MgO$

   D. $Mg + O_2 \rightarrow MgO_2$

9. A gas stove in a home burns methane (natural gas). Balance the equation: $CH_4 + O_2 \rightarrow CO_2 + H_2O$

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10. 24 g of magnesium burns completely with 16 g of oxygen. What mass of magnesium oxide is made?

    A. 8 g

    B. 40 g

    C. 24 g

    D. 48 g

11. An iron nail left outside near the coast turns into rust, and the rust weighs more than the nail did. Has mass been created? Explain.

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12. Photosynthesis is written as 6 CO_2 + 6 H_2O → $C_6H_{12}O_6$ + 6 O_2. Show that this equation is balanced by counting the atoms of each element on each side, and give one reason why changing the temperature or the amount of a reactant can change how fast a reaction goes.

Answer guide for parents

Section 1: Living things

1. B. Mitochondria. Mitochondria release energy from food in cellular respiration. The nucleus holds the DNA and the cell wall supports a plant cell.
2. Any two of: cell wall (supports and protects the cell), chloroplasts (make food by photosynthesis), a large central vacuole (stores water and keeps the cell firm).
3. It controls which substances enter and leave the cell. The cell needs it to take in nutrients and oxygen, remove wastes and keep its inside different from the outside.
4. G C A T T C (C pairs with G, and A pairs with T).
5. C. 20 percent. A pairs with T, so T is also 30 percent and A + T = 60 percent. The other 40 percent is shared equally by C and G, so G = 20 percent.
6. A gene codes for a protein (a chain of amino acids). 300 ÷ 3 = 100 amino acids.
7. A. Two cells, each with the same chromosomes as the original. Mitosis makes two cells that are genetically identical to the original cell.
8. 2 × 2 × 2 × 2 = 16 cells (the number doubles each round).
9. For growth (more cells make the body larger) and to repair or replace damaged and worn-out cells.
10. D. Glucose + oxygen makes carbon dioxide + water + energy released. Respiration uses glucose and oxygen and gives carbon dioxide, water and energy. The first equation is photosynthesis, which makes the glucose and oxygen that respiration uses.
11. A. One quarter. The offspring are TT, Tt, Tt and tt in a 1 : 2 : 1 pattern, and only tt is short, so one quarter.
12. Sweat evaporates from the skin and takes heat away, so the body cools and stays near 37 degrees Celsius. This is homeostasis, kept steady by negative feedback: a rise in temperature causes a response that lowers it.

Section 2: Earth and space

1. D. Geosphere: the sand dunes of the desert. Sand and rock belong to the geosphere. The Hajar mountains are geosphere, mangroves are biosphere and seawater is hydrosphere.
2. They take in carbon dioxide from the atmosphere and water from the sea (hydrosphere), build it into their living tissue (biosphere), and their roots hold the muddy soil in place (geosphere).
3. The Sun heats sea and land water so it evaporates into water vapour. The vapour rises and cools, condenses into clouds, and falls as rain or other precipitation, then flows back to the sea.
4. B. Air over the land rises because land heats faster than water. Land heats faster than water, so the air over land warms and rises. Cooler, denser air from the sea moves in to take its place.
5. C. Summers are usually hot and humid near the coast. Climate is the pattern of weather over many years. The other three statements describe weather on one day.

6. Added by: burning fossil fuels, and respiration by living things (also decay and volcanoes). Removed by: photosynthesis by plants and algae (or by dissolving in the ocean).
7. Gases such as carbon dioxide and methane absorb some of the heat Earth gives off and send part of it back, which keeps the surface warm. Without this natural effect Earth would be too cold for most life.
8. A. It releases extra carbon dioxide, which traps more heat. Burning releases extra carbon dioxide, so more heat is trapped in the atmosphere. The Sun's output does not change because of burning.
9. 5 × 30 = 150 kWh. 150 ÷ 600 = 0.25, so 25 percent.
10. Sunlight keeps arriving every day and cannot be used up, but coal took millions of years to form and cannot be replaced at the rate we use it.
11. B. The water becomes more acidic, so shells and skeletons are harder to build. Dissolved carbon dioxide makes seawater more acidic (ocean acidification), which makes it harder for corals and shellfish to build hard shells and skeletons.
12. As ice melts, more dark water is uncovered and it absorbs more sunlight instead of reflecting it. The extra heat melts more ice. This is a positive feedback loop that increases the change.

Section 3: Matter, forces and energy

1. A. Protons are positive and are found in the nucleus. Protons are positive and sit in the nucleus with the neutrons, which have no charge. Electrons are negative and move around the nucleus.
2. Protons 11, electrons 11 (a neutral atom), neutrons 23 − 11 = 12.
3. C. 7. Valence electrons are the electrons in the outer shell, and the outer shell holds 7.
4. Atoms in the same group have the same number of valence electrons (one each here), and it is the outer electrons that decide how an atom reacts.
5. Their outer shells are full, which is a stable arrangement, so their atoms do not need to gain, lose or share electrons.
6. D. Sodium gives an electron to chlorine, and the ions attract. Sodium gives one electron to chlorine, making Na+ and Cl−. The opposite charges attract and hold the ions together.
7. A covalent bond. In a covalent bond atoms share electrons (usually between non-metal atoms). In an ionic bond electrons are transferred from one atom to another.
8. C. 2 Mg + O2 → 2 MgO. 2 Mg + O2 → 2 MgO has 2 Mg and 2 O on each side. MgO2 is a wrong formula: balancing must change the numbers in front, never the formulas.
9. CH4 + 2 O2 → CO2 + 2 H2O (1 C, 4 H and 4 O on each side).
10. B. 40 g. 24 + 16 = 40 g. This shows the law of conservation of mass: the total mass of the reactants equals the total mass of the products.
11. No. Iron has joined with oxygen (and water) from the air to make rust, so the extra mass came from the oxygen. Atoms are rearranged, not created or destroyed.
12. Left: C 6, H 12, O 12 + 6 = 18. Right: C 6, H 12, O 6 + 12 = 18. They match, so it is balanced. A higher temperature or more reactant means particles collide more often and with more energy, so the reaction goes faster.